DAY 12 | PRE-READ

How AI Writes

A light read before Day 12. About 15 minutes, no code required.

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Why read this?

From today the program adds generative AI: tools like ChatGPT and the OpenAI models behind them, and how to build reliable products with them. Today you learn how these models work, how they are priced, and how to call one from your own code.

1. A very well-read autocomplete

Your phone suggests the next word as you type. A large language model (LLM) does the same thing, but it has learned from an enormous amount of text, so its suggestions can form whole essays, emails or programs. It writes one small piece at a time, called a **token**, each time choosing what is most likely to come next.

2. Paying attention

In "I sat on the bank of the river", you know "bank" means the riverside, not a place for money, because you noticed the word "river". Modern models do this with a mechanism called **attention**: every word looks at the other words to work out what it means in this sentence.

3. Thinkers and sprinters NEW (v4)

Some models answer straight away. Others, called **reasoning models**, think quietly before they answer. Thinking helps with hard problems but takes longer and costs more. You can often choose how hard the model thinks with a setting called **reasoning effort**.

4. Big brains and small brains NEW (v4)

Large models know a lot and reason well, but cost more per question. **Small language models (SLMs)** are cheaper and fast enough to run on a laptop. Many teams use a small model for simple jobs and a large one only when needed.

5. Paying by the word piece NEW (v4)

AI providers charge by the token. What you send is **input**; what the model writes back is **output**, and output usually costs more. A short, focused question is cheaper than pasting a whole document.

Everyday idea	AI term
Word pieces	Tokens
How much it can read at once	Context window
Thinking before answering	Reasoning tokens
Remembering a repeated question	Cached tokens

Key idea: AI can sound confident and still be wrong. Always check important facts.

6. Think about these before class

- Ask an AI assistant the same question twice. Are the answers identical? Why might they differ?
- If every question cost money, how would you write your questions differently?
- Where would a confident but wrong answer be dangerous in a telecom support team?

Kotoba (言葉): words, language